



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

LAB ASSIGNMENT-3

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2023

1. Write a program to implement:

(a) Pointer to an object (b) this pointer Practice both . (dot operator) and -> (arrow operator).

A3_1.cpp

```
1  #include <iostream>
2  using namespace std;
3  class Demo {
4      int x;
5  public:
6      void set(int x) {
7          this->x = x;
8      }
9      void show() {
10         cout << "Value of x = " << x << endl;
11     }
12 };
13 int main() {
14     Demo d;
15     Demo *ptr = &d;
16     ptr->set(10);
17     d.show();
18     return 0;
19 }
20
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS x + v
Value of x = 10
-----
Process exited after 0.2912 seconds with return value 0
Press any key to continue . . . |
```

2. Write a program to swap private values of two classes using a friend function.

```
A3_1.cpp A3_2.cpp
1  #include <iostream>
2  using namespace std;
3  class B;
4  class A {
5      int x;
6  public:
7      void set(int a) { x = a; }
8      void show() { cout << "A.x = " << x << endl; }
9      friend void swap(A &, B &);
10 };
11 class B {
12     int y;
13 public:
14     void set(int b) { y = b; }
15     void show() { cout << "B.y = " << y << endl; }
16     friend void swap(A &, B &);
17 };
18 void swap(A &a, B &b) {
19     int temp = a.x;
20     a.x = b.y;
21     b.y = temp;
22 }
23 int main() {
24     A a;
25     B b;
26     a.set(5);
27     b.set(10);
28     cout << "Before swapping:\n";
29     a.show();
30     b.show();
31     swap(a, b);
32     cout << "\nAfter swapping:\n";
33     a.show();
34     b.show();
35     return 0;
36 }
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS X + v
Before swapping:
A.x = 5
B.y = 10

After swapping:
A.x = 10
B.y = 5

-----
Process exited after 0.2446 seconds with return value 0
Press any key to continue . . . |
```

3. Write a program to add data objects of two different classes using friend functions.

```
A3_1.cpp A3_2.cpp A3_3.cpp
1  #include <iostream>
2  using namespace std;
3  class B;
4  class A {
5      int x;
6  public:
7      A(int a) { x = a; }
8      friend int add(A, B);
9  };
10 class B {
11     int y;
12 public:
13     B(int b) { y = b; }
14     friend int add(A, B);
15 };
16 int add(A a, B b) {
17     return a.x + b.y;
18 }
19 int main() {
20     A a(10);
21     B b(20);
22     cout << "Sum = " << add(a, b);
23     return 0;
24 }
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS X + v
Sum = 30
-----
Process exited after 0.2465 seconds with return value 0
Press any key to continue . . . |
```

4. Write a program to demonstrate the working of a friend class.

```
A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp
1  #include <iostream>
2  using namespace std;
3  class First {
4      int x = 10;
5      friend class Second;
6  };
7  class Second {
8  public:
9      void show(First f) {
10         cout << "Value of x = " << f.x;
11     }
12 };
13 int main() {
14     First f;
15     Second s;
16     s.show(f);
17     return 0;
18 }
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS X + v
Value of x = 10
-----
Process exited after 0.26 seconds with return value 0
Press any key to continue . . . |
```

5. Write a program using Array of Objects to display the area of multiple rectangles.

```
A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp A3_5.cpp
1  #include <iostream>
2  using namespace std;
3  class Rectangle {
4      int l, b;
5  public:
6      void set() {
7          cin >> l >> b;
8      }
9      void area() {
10         cout << "Area = " << l * b << endl;
11     }
12 };
13 int main() {
14     Rectangle r[3];
15     for (int i = 0; i < 3; i++) {
16         cout << "Enter length and breadth: ";
17         r[i].set();
18     }
19     for (int i = 0; i < 3; i++) {
20         r[i].area();
21     }
22     return 0;
23 }
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS x + v
Enter length and breadth: 45
1
Enter length and breadth: 45
2
Enter length and breadth: 45
3
Area = 45
Area = 90
Area = 135

-----
Process exited after 12.19 seconds with return value 0
Press any key to continue . . . |
```

6. Write a program to define function cube() as inline for calculating cube of a number.

```
A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp A3_5.cpp A3_6.cpp
1  #include <iostream>
2  using namespace std;
3  inline int cube(int x) {
4      return x * x * x;
5  }
6  int main() {
7      int n;
8      cout << "Enter a number: ";
9      cin >> n;
10     cout << "Cube = " << cube(n);
11     return 0;
12 }
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS x + v
Enter a number: 4
Cube = 64
-----
Process exited after 3.349 seconds with return value 0
Press any key to continue . . .
```

7. Write a program to pass an object as an argument and return the object from a function.(a) Use pass-by-value (b) Use pass-by-address.

```
A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp A3_5.cpp A3_6.cpp A3_7a.cpp
1  #include <iostream>
2  using namespace std;
3  class Test {
4  public:
5      int x;
6  };
7  Test change(Test t) {
8      t.x = 20;
9      return t;
10 }
11 int main() {
12     Test t1, t2;
13     t1.x = 10;
14     t2 = change(t1);
15     cout << "Returned value = " << t2.x;
16     return 0;
17 }
```


OUTPUT:

```
C:\Users\DELL\Desktop\OOPS X + v
Returned value = 20
-----
Process exited after 0.2769 seconds with return value 0
Press any key to continue . . . |
```

```
A3_1.cpp A3_2.cpp A3_3.cpp A3_4.cpp A3_5.cpp A3_6.cpp A3_7a.cpp A3_7b.cpp
1  #include <iostream>
2  using namespace std;
3  class Test {
4  public:
5      int x;
6  };
7  Test change(Test *t) {
8      t->x = 30;
9      return *t;
10 }
11 int main() {
12     Test t1, t2;
13     t1.x = 10;
14     t2 = change(&t1);
15     cout << "Returned value = " << t2.x;
16     return 0;
17 }
```

OUTPUT:

```
C:\Users\DELL\Desktop\OOPS X + v
Returned value = 30
-----
Process exited after 0.2881 seconds with return value 0
Press any key to continue . . . |
```